Conversation folding in llm-gateway

A report of the work done in one session, 22–23 October 2026: from folding receipts to a gateway-owned fact ledger. Every number here comes from the benchmark result files committed in the repository.

In short. Folding replaces old conversation turns with a short summary so each request sends fewer tokens. Today it gained receipts, a benchmark against real models, and a design that keeps facts correct when they change. **On stable facts it is proven:** every fact kept, at a 47–54% saving even after prompt caching. **On changing facts it is now safe but not yet economical:** with a gateway-owned fact ledger, no folded answer to a changed fact was wrong in the final rounds, because the gateway refuses any summary it cannot verify; with the cheap summarizer (gpt-4.1-nano) that meant almost no folds and no saving. 35 benchmark runs, 4,099 paired calls, about $7 at provider prices.

1. Folding receipts

The day began by writing two plans for folding and finishing the receipts plan after its four decisions (D1–D4) were approved. It was built in three phases, one commit each, and merged as **PR 53**:

- the summary the model was sent comes back to the caller in `x-gateway-fold-summary`, and the main receipt binds the folded request (`folded_request_digest`);
- the summarizer call gets its own allow and outcome receipt, so a fold is evidence in its own right;
- background folding is refused for receipts tenants, which have no caller to hand the salt to;
- a request refused after its fold still returns the salt and fold headers, so the caller can check what it paid for.

A third reviewer's findings followed in **PR 55**: a tenant's own `allowed_models` is enforced at runtime (and for its summarizer), refusals after a fold return the salt, and the benchmark's HTTP client no longer follows redirects, so an API key cannot be sent elsewhere.

2. A benchmark against real models

`tools/fold_retention.py` sends every call twice, to a tenant without folding and one with it, same model, same messages, temperature 0, and asks for ten facts that the conversation states, corrects and confuses with a second project. Answers are scored by strict typed equality; a fact counts only if it was folded away and the unfolded tenant got it right. Groq's daily token limit stopped the first runs, so the benchmark moved to OpenRouter (gpt-4.1-mini answering, gpt-4.1-nano summarizing): **PR 54**.

3. Does folding still pay with prompt caching?

The gateway bills every prompt token at full price, but OpenAI serves a repeated prefix from its cache at a quarter of it, and an unfolded conversation, which only grows at its end, gets most of that discount. The harness learned to record cached tokens, price them at the cached rate, alternate which tenant goes first, and run each profile twice (**PR 57**, with a dashboard and the first PDF report):

Run	Facts kept	Cached share, unfolded / folded	Cost with caching	Saving
fixed, 200 calls	1492 / 1492	59% / 19%	$0.259 → $0.120	54%
fixed, repeat	1492 / 1492	64% / 20%	$0.243 → $0.118	51%
mutating, 120 calls	772 / 816	57% / 15%	$0.136 → $0.066	52%
mutating, repeat	836 / 867	65% / 5%	$0.123 → $0.060	51%

Caching does not erase the saving. But both mutating runs lost the latest *owner* right after a fold and never got it back, and that started the rest of the day.

4. Chasing the lost owner

First, diagnostics (**PR 58**): the result files now keep what each tenant answered when wrong, every summary the folded tenant was sent, each call's fold outcomes, and which checks a rejected summary failed. The first summaries showed the cause at once: a chronological story that stated the old owner first, the correction later, and was cut at 1,200 characters exactly where the newest corrections sat. Then one safeguard per round, each two mutating runs of 120 calls:

Round	Facts kept	Saving	What it showed
Current-facts block and field/value pins	667/667, 705/818	42%, 45%	The summarizer read “Noor replaces Mira” backwards

Round	Facts kept	Saving	What it showed
A replaced value may not be current	640/647, 695/695	36%, 47%	A stated-once port was dropped
Bare numbers pinned; rejection reasons	452/452, 727/727	22%, 37%	Old values became mandatory; folding stalled
Replaced values not required	488/488, 510/510	27%, 31%	The first fold took 8 attempts
One corrective retry	793/811, 662/749	48%, 46%	Folding flowed; two lines for one fact
Same, gpt-4.1-mini summarizing	715/715, 720/720	43%, 40%	Clean, but few folds at its price

Every safeguard closed the failure it targeted and exposed the next. Strict checks lost nothing but stopped folding; looser ones let the summarizer's mistakes through. Two reviews pasted during the day named the root: the gateway was inferring each fact's current value from the summary's wording.

5. The fact ledger

PR 59 moves the facts out of the summary. The summarizer now returns JSON: fact updates from the new turns, each a field, a value and the exact sentence it came from, plus a short context note. The gateway keeps the facts per conversation and writes the `Current facts` block itself. An update is applied only if:

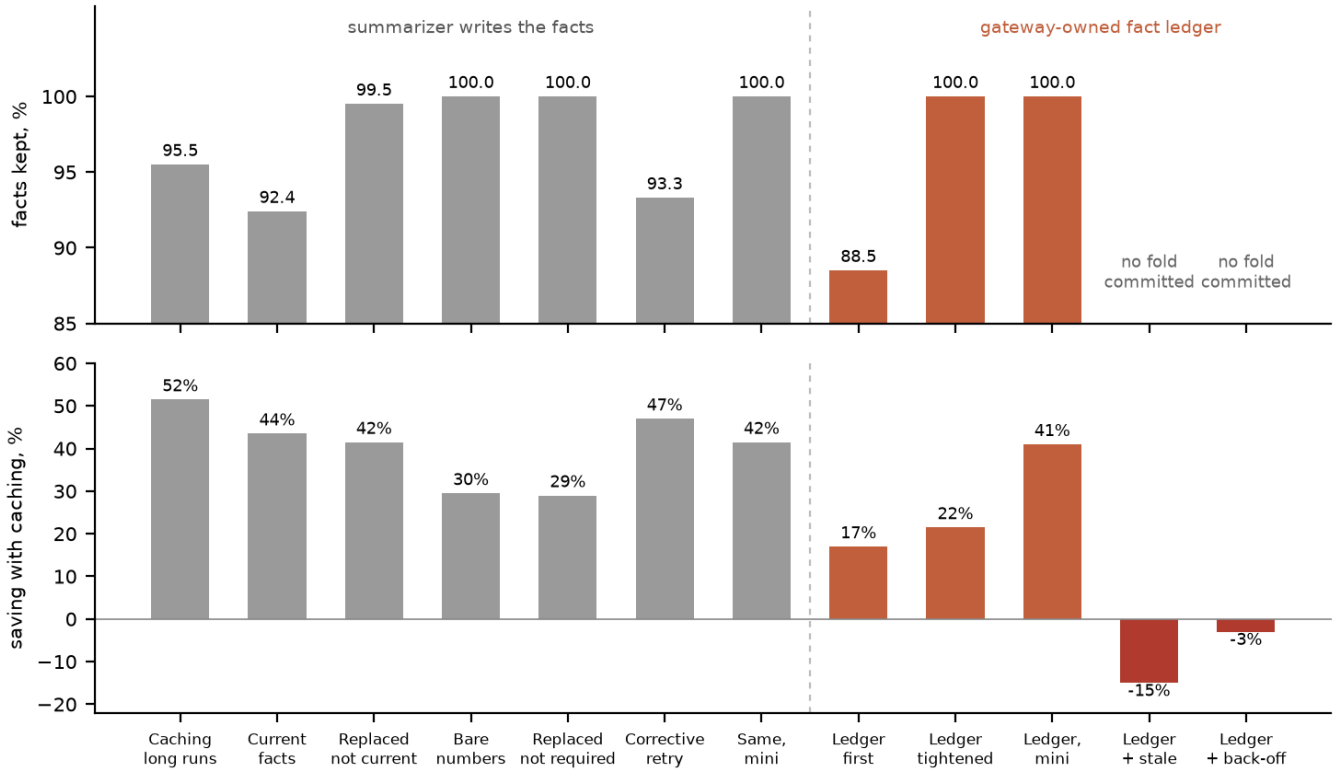
- its evidence is one sentence from the new turns, holds the value, and names the fact;
- its value is not the one its own sentence replaces (“Noor replaces Mira” can never set Mira);
- it is not a second name for a kept fact, and goes to the kept fact its sentence names;
- afterwards, no kept fact still holds a value the new turns replace.

Updates apply in the conversation's order, not the reply's. Each fact keeps its source sentence, turn and last eight replaced values. A later fold only adds updates, so it cannot drop or flip a fact. One failed update fails the whole reply, which gets one corrective retry; the back-off after a rejection doubles each time, so a conversation that cannot be folded stops paying for attempts.

Ledger version	Run	Facts kept	Folds	Saving
First	fixed, 200 calls	1441 / 1441	5 of 16	54%
First	mutating ×2	625 / 706; none	4 of 10; 0 of 15	49%; –15%
Evidence names its fact	mutating ×2	417 / 417, 421 / 421	1 of 14, 1 of 13	22%, 21%
Same, gpt-4.1-mini	mutating	710 / 710	2 of 2	41%
Same	fixed, 200 calls	1223 / 1347	7 of 14	55%
+ stale facts refused	fixed, 200 calls	1417 / 1417	4 of 18	47%
Same	mutating ×2	none	0 of 15, 0 of 15	–14%, –16%
+ growing back-off	mutating ×2	none	0 of 4, 0 of 4	–3%, –3%

The first ledger let two kinds of wrong data in: a value taken from a turn that stated two facts, and one fact under two names. The tightened ledger then let a correction start a new fact beside the old one (*ceiling: 1800* beside *spending ceiling usd: 2400*). Each became a rule with its own test.

Every round on the mutating scenario



Facts kept counts only facts folded away. Grey: the summarizer writes the facts. Orange: the gateway keeps them. Red: the folded tenant cost more than the unfolded one.

6. How the work was tested

- The full Rust suite (580 tests across 11 test binaries) and clippy with warnings as errors ran after every change; the Python tools have 57 tests.
- Every new rule was broken on purpose, one at a time, to confirm a test fails: 91 deliberate breaks across the diagnostics, safeguards and ledger. Twice a break went unnoticed; both times the gap was closed with a test.
- Before spending on a run, new checks were replayed against real summaries from earlier runs, and the benchmark's own sentences were run through the ledger, to catch false rejections.
- Result files were checked for summed spends, the scoring rule, both call orders, and the absence of key-like strings. CI passed on Linux and Windows for every merged PR and for PR 58.

7. What is established, and what is not

Established	Not yet established
Folding saves 47–54% on stable facts, prompt caching included, with every fact kept.	That a conversation full of corrections saves money: with gpt-4.1-nano almost every fold was refused.
With the ledger, the summarizer's mistakes become refused folds, not wrong answers, in every final run.	A correction followed by a re-fold, in a real run: no ledger run reached the fold cap with a later correction.
gpt-4.1-mini extracts facts cleanly (no refusals in its run).	That mini pays off: at its price the cost check let 2 folds through in 120 calls.
A conversation that cannot be folded now costs about 3% extra, not 15%.	Stable results over many repeats: most rounds were two runs.

8. Open items

- **Merge PR 58, then PR 59** (stacked). PR 58's CI passed on every check; PR 59's CI had not yet run.
- **x-request-id.** The middleware replaces the handler's id, so a response's id is not the one in the logs or receipts. A separate defect; the integration tests already note it.
- **The cost check for a costlier summarizer.** Mini needs a fold to pay off over more requests than the lifecycle profile's four; worth measuring on the economics profile.
- **A longer mutating run** that reaches the fold cap and then re-folds after a correction.

- **The dashboard** built for PR 57 must be shared from its Share menu before others can open it.

Appendix: pull requests

PR	What	State
53	Folding receipts: summary header, folded-request digest, fold allow and outcome	merged
54	Fold-retention benchmark on OpenRouter	merged
55	Reviewer fixes: tenant allowlist at runtime, salt after a refused fold, no redirects	merged
57	Prompt-caching costs, alternating arms, repeated long runs, dashboard and first report	merged
58	Diagnostics, safeguards for changing facts, and documented limits	open, CI green
59	The gateway-owned fact ledger and a growing back-off	open

Benchmark totals for the session: 35 runs (32 OpenRouter, 3 Groq), 4,099 paired calls. Provider cost with caching for the 29 runs that recorded it: $6.96. Gateway-billed spend for both tenants, at full prompt price: $19.54.